

Ashish Chandra, Ph.D.

Assistant Professor

Department of Management and Quantitative Methods

College of Business, Illinois State University

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ACADEMIC APPOINTMENT

Assistant Professor - Management & Quantitative Methods

Aug 2022 – Present

College of Business, Illinois State University

EDUCATION

Ph.D.	Purdue University, Krannert School of Management Now: Mitch Daniels School of Business Ph.D. in Management: Specialization in Quantitative Methods Minor in Electrical & Computer Engineering Title: Techniques for Uncertainty Quantification, Risk Minimization, with Applications to Risk-Averse Decision Making Advisor: Mohit Tawarmalani	July 2022
M.S.	Indian Institute of Technology Bombay (IIT Bombay) Department of Industrial Engineering and Operations Research Master of Science in Operations Research Title: Systemic Risk in Financial Networks Advisor: Veeraruna Kavitha	Dec 2015
B.S.	University of Mumbai - DAE Centre for Excellence in Basic Sciences Bachelor of Science in Mathematics	July 2013

RESEARCH INTEREST

Interests:	Network/System Reliability Problems, Robust Network Design, Transportation and Supply Chain Problems, Uncertainty Quantification, Contagion in Financial Networks, and Kidney Exchange problem.
Methodologies:	Mathematical Programming, Stochastic Optimization, Decision Making under Uncertainty, Algorithm Design, Applied Probability, and Statistics.

TEACHING INTERESTS

Operations Management, Quantitative Methods/Analysis, Deterministic and Stochastic Operations Research, Mathematical Programming, Convex Optimization, Applied Probability and Statistics, Optimization Models.

JOURNAL PUBLICATIONS

- Ashish Chandra JIMO
- Ashish Chandra OR Letters

- **Ashish Chandra** OR Forum
- **Ashish Chandra** and Jong Gwang Kim. “Reliability certification of supply chain networks under uncertain failures and demand.” *Annals of Operations Research (2025): 1-31*.
- **Ashish Chandra** and M. Tawarmalani. “Probability estimation via policy restrictions, convexification, and approximate sampling.” *Mathematical Programming 196.1 (2022): 309-345*.
- Y. Chang, C. Jiang, **Ashish Chandra**, S. Rao, and M. Tawarmalani, “Lancet: Better Network Resilience by Designing for Pruned Failure Sets”, *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, vol. 3, no. 3, pp. 1–26, 2019.

CONFERENCE PROCEEDINGS

- **Ashish Chandra**. “MidWest DSI, 2026
- **Ashish Chandra**. “IEOM, 2026
- **Ashish Chandra**. “DSI pedagogy, 2026
- **Ashish Chandra**. “Quantifying Worst-Case Supply Chain Carbon Footprints: A Network-Based Approach”, *10th North American Conference on Industrial Engineering and Operations Management, IEOM 2025*.
- **Ashish Chandra**, “Computing Inefficiency of Service Networks under Random Failures”, *Proceedings of the IISE Annual Conference and Expo 2024*.
- **Ashish Chandra**, “Robust Optimization for Cost Validation in Designing Service Networks”, *7th European Conference on Industrial Engineering and Operations Management, IEOM 2024*.

WORKING PAPERS

- Jong Gwang Kim, Ashish Chandra, Abolfazl Hashemi, Christopher Brinton. “A Fast Single-Loop Primal-Dual Algorithm for Non-Convex Functional Constrained Optimization”.
- “Importance Sampling Techniques for Probabilistic Certification of Network Performances”.
- “Multi-objective optimization for resilient and robust network design”.
- “First-Order Primal-Dual Method for Linearly Constrained Nonconvex Optimization”, with Jong Gwang Kim.
- Ashish Chandra, Jong Gwang Kim. “Reliability certification of supply chain networks under uncertain failures and demand.”
- Ashish Chandra, Mohit Tawarmalani. “Minimizing risk measures with applications in network traffic engineering”.

CONFERENCE AND SYMPOSIUMS

- **A. Chandra**, “Quantifying Worstcase Supply Chain Carbon Footprints: A Network-Based Approach”.
 - 10th North American Conference on Industrial Engineering and Operations Management, 2025
- **A. Chandra**, “Computing Inefficiency of Service Networks under Random Failures”.
 - IISE Annual Conference & Expo 2024
- **A. Chandra**, “Robust Optimization for Cost Validation in Designing Service Networks”.
 - 7th European Industrial Engineering and Operations Management Conference, 2024
- **A. Chandra and M. Tawarmalani**, “Risk Minimization For Network Traffic Engineering”.

- INFORMS Annual Meeting 2023
- **A. Chandra, “Risk minimization in supply chain networks”.**
 - Decision Science Institute - Annual Conference 2023
- **A. Chandra and M. Tawarmalani, “Convex approximations of risk measures & applications to chance-constrained programming”.**
 - INFORMS Annual Meeting 2022
 - International Conference on Continuous Optimization (ICCOPT), 2022
- **A. Chandra and M. Tawarmalani, “Optimization Techniques for Probability Estimation and Applications to Network Reliability Problems”.**
 - INFORMS Annual Meeting 2021
 - SIAM Annual Meeting 2021
 - Institute of Industrial and Systems Engineers - Annual Conference and Expo 2021
 - Krannert Doctoral Research Symposium 2019, 2020 & 2021
 - INFORMS Optimization Society Conference 2020 - Postponed due to Covid-19
 - CRM/DIMACS Workshop on Mixed-Integer Nonlinear Programming 2019
- **Y. Chang, C. Jiang, A. Chandra, S. Rao, and M. Tawarmalani, “Lancet: Better Network Resilience by Designing for Pruned Failure Sets”.** (Krannert Doctoral Research Symposium 2017, 2018)
- **Session Chair**
 - “Supply Chain Resilience and Sustainability” - MidWest DSI Annual Meeting 2025.
 - “Supply Chain Management” - IISE Annual Conference and Expo 2024.
 - “SUPPLY-CHAIN RISK and RESILIENCY: Supply Chain Risk Mitigation” - DSI Annual Meeting 2023.
 - “Machine Learning and Optimization Applications” - INFORMS Annual Meeting 2023.
 - “Teaching and Learning Symposium - 2023”, Illinois State University.
 - “Optimization techniques for decision making under risk” - INFORMS Annual Meeting 2022.
 - “Nonlinear Optimization Techniques in Stochastic Optimization”- INFORMS Annual Meeting 2021.

TEACHING EXPERIENCE

- **Instructor: Illinois State University** *(1 - Highest rating and 5 - Lowest rating)*
 - **MQM 100 - Statistical Reasoning**
 - Summer 2025 - **1.70**
 - Summer 2024 - **Rating: 1.56**
 - Fall 2023 - **Rating: 1.54**
 - Spring 2023 - **Rating: 1.12**
 - Fall 2022 - **Rating: 1.28**
 - **MQM 227 - Operations Management**
 - Winter 2025 - **Rating: -**
 - Fall 2025 - **Rating: 1.47**
 - Spring 2025 - **Rating: 1.94**
 - Fall 2024 - **Rating: 1.66**
 - Spring 2024 - **Rating: 1.38**

Fall 2023 - **Rating: 1.65**
 Spring 2023 - **Rating: 2.24**

- **MBA 416 - Applied Tools For Business Decision Making**

Spring 2025 - **Rating: 1.06**

Fall, 2025, Fall 2024, Spring 2024

- **Purdue University - Instructor**

- **MGMT 305 - Business Statistics:** Core course for BS in Business Analytics and Information Management.
 Summer 2019 - Krannert Certificate for Distinguished Teaching.
 Responsibilities: Deliver teaching lectures (130 minutes daily), hold tutorial sessions, and grade homework.
 Lectures also included an introduction and illustrations on Minitab.

- **Purdue University - Teaching Assistant**

- **MGMT 670 - Business Analytics:** Core course for MS in Business Analytics and Information Management.
 Mohit Tawarmalani - Spring 2021 & Fall 2021
 Responsibilities: Hold weekly tutorial sessions and grade homework.
- **MGMT 306 - Management Science:** Core course for BS in Business Analytics and Information Management.
 Anureet Saxena - Fall 2016, Andy Alexander - Spring 2017, Fall 2017, Spring 2018
 Responsibilities: Deliver weekly hour-long recitation lectures, design exams, and hold weekly tutorial sessions.
 Lectures included illustrations for optimization modeling and simulation on Microsoft Excel.
- **MGMT 305 - Business Statistics**
 Gary Evans - Fall 2018
 Responsibilities: Deliver weekly hour-long recitation lectures, and hold weekly tutorial sessions.

- **Industrial Engineering & Operations Research - IIT Bombay - Teaching Assistant**

- **IE 611 - Introduction to Stochastic Models**
 Veeraruna Kavitha - Spring 2016
 Responsibilities: Design and Grade homeworks and exams.

HONORS AND AWARDS

- Instructional Innovation Grant
 Center for Integrated Professional Development, Illinois State University 2025
- Winner of the IEOM Supply Chain and Logistics Competition
 7th European Industrial Engineering and Operations Management Conference Jul 2024
- University Research Grant, Illinois State University 2024 – 2025
- University Research Grant, Illinois State University 2023 – 2024
- Givens Associate Fellowship, Argonne National Laboratory, Illinois, USA Jun – Aug 2021
- Graduate Research Assistantship, Purdue University 2016 – Jul 2022
- Best Paper Runner-up, Krannert Doctoral Research Symposium, Purdue University 2020, 2021
- Purdue Graduate Student Travel Grant, Purdue University 2020, 2022
- Krannert Certificate for Distinguished Teaching (Highest teaching award for Krannert instructors) Summer 2019
- Charpak Research Scholarship, French Government May – Jul 2015
- Summer Research Fellowships:
 National Program on Differential Equations - Theory, Computation & Applications, India May – Jun 2013
 Indian Academy of Sciences, India May – Jun 2012
- Innovation in Science Pursuit for Inspired Research - INSPIRE Fellowship, Government of India 2010 – 2013

PROGRAMMING & SOFTWARE SKILLS

- Programming: Python (proficient), Matlab, C/C++
- Optimization Software: Gurobi, AMPL, GAMS, CPLEX, MOSEK
- Statistics: Minitab, Matlab, R, Mathematica
- Database: SQL, MySQL
- Machine Learning: Tensorflow, Keras
- Others: AnyLogic, Optimization Modeling - Risk analysis & Simulation in Microsoft Excel

OTHER RESEARCH EXPERIENCES

- **Argonne National Laboratory - Mathematics & Computer Science Division**
Givens Associate. Host: Anirudh Subramanyam and Vishwas Rao Jun – Aug 2021
Topic: Importance Sampling for Probabilistic Certification of Network Performances.
- **Indian Institute of Technology Bombay - Industrial Engineering & Operations Research**
Research Assistant. Host: Veeraruna Kavitha Jan – May 2016
Topic: Brownian Motion on One Dimensional Grid with Random Deletions.
- **French Institute for Research in Computer Science and Automation, Sophia Antipolis, France**
Summer Research Internship. Host: Konstantin Avratchenkov May – Jul 2015
Topic: Random Walks with Restarts & Applications in Non Convex Optimization.
- **Indian Institute of Technology Bombay - Industrial Engineering & Operations Research**
Summer Research Internship. Host: K.S. Mallikarjuna Rao May – Jul 2013
Topic: Nash Equilibrium in Distributed Systems and Applications in Road Traffic.
- **Indian Institute of Science Bengaluru - Department of Mathematics**
Summer Research Internship. Host: Manjunath Krishnapur May – Jul 2012
Topic: Random Walks and Electric Networks.

PEER REVIEW SERVICES

- Year: 2025
 - Reviewer for the Journal “Mathematical Programming Computation”.
 - Reviewer for “Supply Chain Resilience and Sustainability”, Mid-West Decision Science Institute Conference, 2025.
 - Reviewer for the 2025 Academy of Management Annual Conference.
- Year: 2024
 - Reviewer for IISE Annual Conference 2024.
 - Reviewer for the Journal “Mathematical Programming Computation”.
 - Track chair for “Supply chain management and Logistics”, Mid-West Decision Science Institute Conference, 2024.
 - Reviewer for the 2024 Academy of Management Annual Conference.

CERTIFICATION & PROFESSIONAL MEMBERSHIPS

- Coursera Certification:
 - Data Science Specialization**, a 4-course specialization by IBM on Coursera.
 - Deep Learning Specialization**, a 5-course specialization by DeepLearning.AI on Coursera.
- Professional Memberships:
 - Institute for Operations Research and the Management Sciences (INFORMS)
 - Institute of Industrial and Systems Engineers (IISE)
 - Society for Industrial and Applied Mathematics (SIAM)

MENTORSHIP

- Student Companion Program, IIT Bombay Aug 2014 – Apr 2015
Assisted 5, first year Master students in their academic, non-academic issues and in their overall development.
- Student Mentorship Program, University of Mumbai Aug 2012 – Apr 2013
Mentored undergraduate freshmen students from diverse socio-economic backgrounds.

NON-ACADEMIC INTERESTS

Cricket - Purdue Cricket Club, Improv theatre, Marathon running, SKY Purdue - Yoga & Meditation.